

The Elements of DeFi

DeFi does not have an innovation problem. It has a vocabulary problem. Nearly every protocol ever shipped is a recombination of about fifty recurring mechanisms — and once you can name them, most of the industry stops looking novel and starts looking legible.

Call it a periodic table if you like, but it is not one: no periodic law, no atomic number, nothing recurring down a column. Chemistry discovered its elements. These were designed, in a hurry, by people shipping to production — which is exactly why the combinations are the part worth writing down.

THEME

Dark

Light

v1.0 · reviewed 2026-08-04

48 core · 10 candidate · 10 contested · 29 laws · 19 hazards

READ A PROTOCOL

Pick one. It lights up in the table below, numbered in the order each part became necessary. Four of these are dead — and those are the instructive ones.

Aave v3 Pooled lending 10 elements	Uniswap v3 Spot exchange 4 elements	Maker / Sky CDP stablecoin 10 elements	Liquity v1 CDP stablecoin 6 elements	GMX v2 Perpetual futures 8 elements	Lido v2 Liquid staking 5 elements
CoW Protocol Intent exchange 4 elements	CCTP v2 Fast Cross-domain transfer 4 elements	Centrifuge Real-world credit 8 elements	Terra / Anchor + Algorithmic stablecoin 6 elements	Mango Markets + Perpetual futures 6 elements	Euler v1 + Pooled lending 7 elements

Rows are how much must already exist
S0 needs nothing but the ledger. S4 needs other people to agree. Deeper is not more dangerous — it is more dependent, and confusing those two is how you misprice a protocol.

Blocks are families of substitutes
Everything in a block answers one question. Swap Cp for Cl and you still have a DEX. That is what a block means and the only thing it means.

The shape is the finding
Families cluster at one depth — credit is all S3, pricing all S1. And S3 is enormous: most of DeFi is machinery for owing, pricing and unwinding obligations.

Colour only ever means hazard
There is exactly one coloured thing in this interface and it is danger. Everything else earns its distinction from position, shape and lightness.

VIEW

The table

Laws

Hazards

ARRANGEMENT

By depth

By family

DIMENSION

Flat

3D

READING ORDER

By group

By stratum

By ID

SEARCH

symbol, name or ID

Row = prerequisite depth, exactly. **Blocks** inside a row are families of substitutes. Order within a block is **not** semantic.

Vl
Staking & validator...

candidate

In
Intent & solver execution

core

Xm
Cross-domain message...

core

Tg
Delayed governance execution

core

Xf
Cross-domain asset transfer

core

Up
Putable implementation...

core

Rl
Resource lock / reservation

candidate

Gp
Guardian or pause

core

Of
Optimistic fill & pre-emptment

candidate

Au
Delegated execution scope

core

Rs
Restaking / shared security

candidate

Contested register – 10 entries

Held outside the table because a necessary criterion is genuinely disputed.
Not usable in a formula without a note. The separation means “argued about”, not “filed away”.

P001

Bc

Bonding-curve issuance

n

contested

P002

Tw

Time-weighted AMM execution

n

contested

P003

Da

Dutch-auction descent

n

contested

P004

Cg

Credit delegation

n

contested

P005 Ir Insurance reserve fund	P006 Zk Verifiable state proof	P007 Ve Vote-escrow allocation	P008 Kg Credential-gated transfer
 contested	 contested	 contested	 contested
P009 Ua Unified-balance ledger	P010 Sq Shared ordering commitment		
 contested	 contested		

HOW TO THINK WITH IT

Six questions, in this order. Run them against anything — a protocol you are auditing, a pitch deck, your own design — and the table stops being a diagram and becomes a method. Question three pays for the other five.

1 What are its parts?

Name every recurring financial state transition and refuse everything else. ERC-4626 is a socket, not a mechanism. A hook is somewhere to put an element, not an element. “Points” is marketing. Most whitepapers describe three real mechanisms and forty pages of scenery.

4 Does anything feed itself?

Reflexive collateral is the most reliable way to lose everything ever deployed on-chain. A token that is simultaneously the collateral, the market that prices it, and the capital backing it has three defences that all fail on the same news. This is the one shape that goes to zero rather than merely down.

2 What did each part force?

Parts are not chosen freely — each one drags in the next. Leverage forces a truth source, a solvency test and a way for the position to end. A protocol with leverage and only two of those three is not innovating. It is missing a part, and you have just found it.

5 Who eats the loss?

For every fee, delay, slippage and advance, name the claim class or the underwriter. An unassigned loss has not vanished — somebody is carrying it without being paid to, and usually it is the depositor who never read this far. If the documentation cannot name them, that is the answer.

3 Where does it read the outside world?

This is the question that pays. Every catastrophic loss this table explains runs through the price, not the code. Ask what it would cost to move that input — measured against pool depth, never against the attacker’s balance — and if the answer is less than what can be borrowed against it, the protocol is already lost.

6 What did the formula fail to say?

The residue, and it is the most useful line on the page. For a lending market it is small. For anything touching real-world assets it is larger than the formula and contains the actual risk. A decomposition that reports no residue was not finished.

LEGEND

Stratum S0 – S4



Prerequisite depth, not risk. Carried by the band, the S-token and lightness — never hue.

-  core
-  candidate: recurrence evidence short

contested: note required



- / degenerate limit: not an element
-  no denominator – failures counted, survivors never were

-  hazard: forbidden
-  hazard: elevated
-  hazard: unverifiable

Hazard membership is categorical. It is never a probability, because no exposure or survivor counts exist.

- i– Interface – Can these exchange the required calls, tokens, messages, proofs?
- e– Economic – Do combined incentives, liquidity and loss allocation stay solvent?
- t– Trust – Who can attest, censor, upgrade, pause, mint, freeze?
- n– Informational – Who observes the order before it settles?

This table explains roughly two thirds of DeFi incidents by count and a minority of them by dollars. That is not a defect in the table — it is the most useful finding on this page. Your largest risk is not your financial design. It is your keys, your signing interface and your build pipeline. Get the mechanism right and you have eliminated the failures people write papers about, not the ones that actually took the money.